

STRATUM

Owner's Manual

Alpha v10 · for macOS 12 or later

Everything in STRATUM, where to find it, what it does, and how it compares with the music software you may already use.

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1 Welcome and installing

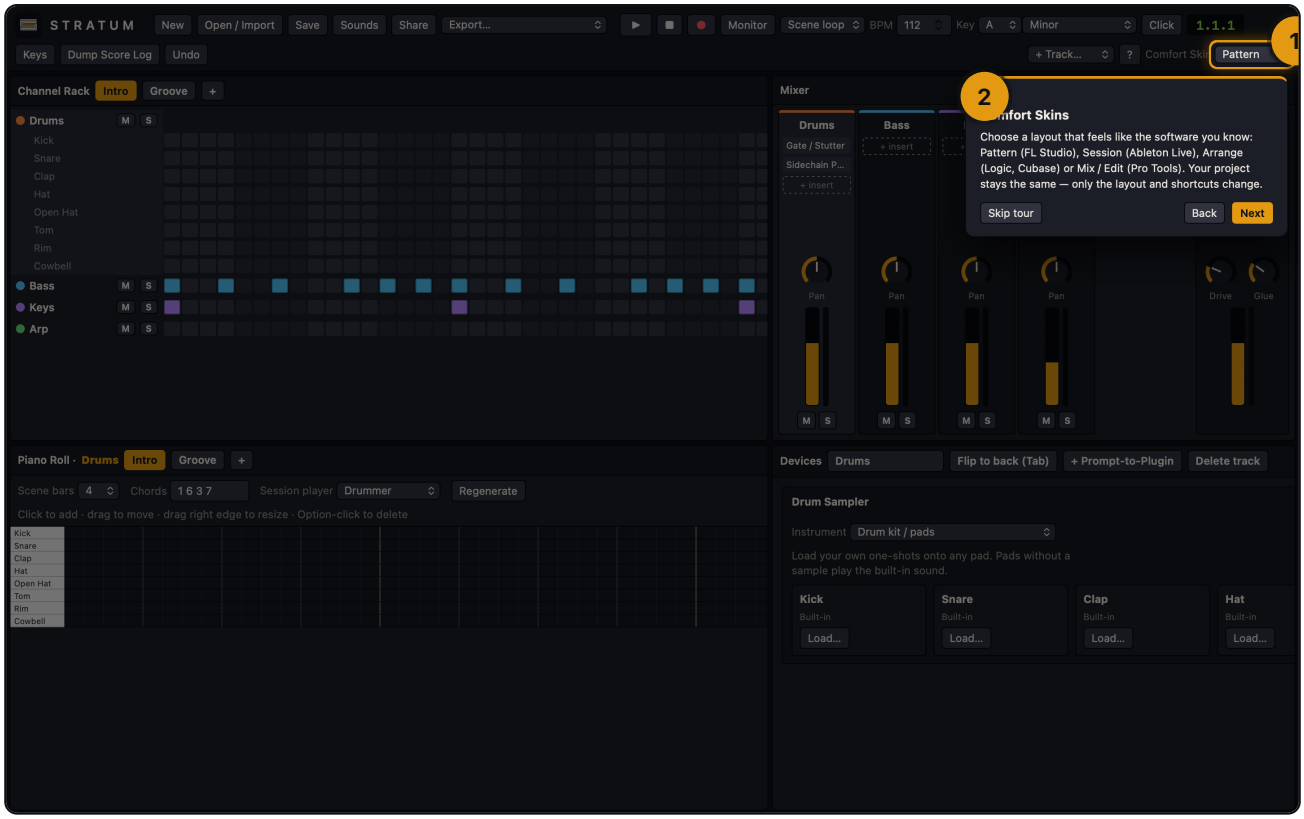
STRATUM is a music production app for the Mac. It plays and records audio, runs software instruments and effects — its own, and the Audio Unit, VST3 and VST2 plugins installed on your Mac — and opens projects from FL Studio, Ableton Live, Logic Pro and Reason.

Installing

1. Open **STRATUM.dmg** and drag **STRATUM** into **Applications**.
2. Open STRATUM. Because this alpha is not yet notarised by Apple, macOS stops it the first time.
3. Open **System Settings › Privacy & Security**, scroll down and click **Open Anyway** next to STRATUM. You only do this once.

The first time you open it

STRATUM opens with a short demo song and asks whether you want a tour. The tour highlights each part of the window and lets you try Play and the sound library as you go. You can take it again any time from **Help › Take the Tour**.



The guided tour pointing out the Comfort Skin menu.

1 · Highlighted control — the part of the window the tour is explaining.

2 · Tour card — what it does; Next and Back move through the tour, Skip leaves it.

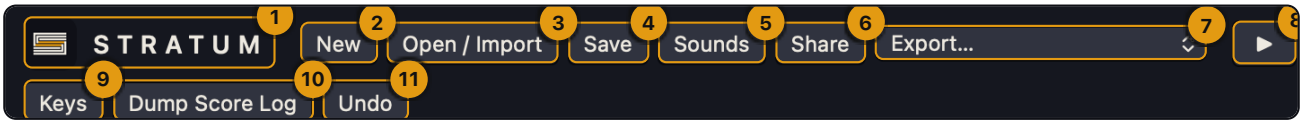
Permissions STRATUM may ask for

- **Microphone** — only when you first record.

- **Photos** — only when you share to an Instagram Story, so the video can reach your phone.
- **Camera** — only when you go live with your camera on.
- **Screen & System Audio Recording** — when you record from another app, or go live, so viewers hear your sound.

2 The window at a glance

The window is a bar of controls along the top and a set of panels below. Which panels you see, and where, depends on the Comfort Skin (Chapter 3); this is the **Pattern** skin.



The top bar, left half.

1 · STRATUM — the app name; the window title shows your project's name.

2 · New — starts an empty project.

3 · Open / Import — opens STRATUM projects and imports other software's projects, MIDI and audio (Chapter 4).

4 · Save — saves a .stratum project file, with all its audio inside it.

5 · Sounds — the built-in sound library (Chapter 11).

6 · Share — makes a branded video and audio file and sends or posts it (Chapter 14).

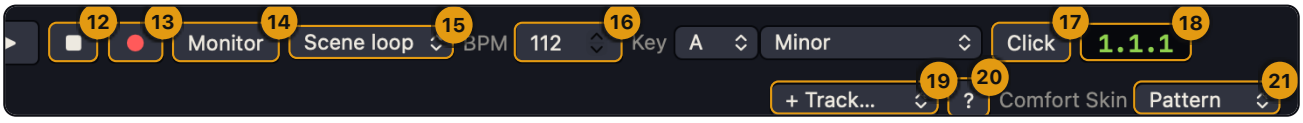
7 · Export — WAV mixdown, MIDI file or DAWproject.

8 · Play / stop — Space bar does the same.

9 · Keys — play the selected track from the computer keyboard.

10 · Dump Score Log — puts what you played in the last five minutes into the selected clip — even if you weren't recording.

11 · Undo — ⌘Z.



The top bar, right half.

12 · Stop — stops and returns to the start.

13 · Record — records onto the armed audio track (Chapter 9).

14 · Monitor — hear your input while recording.

15 · Scene loop / Song — loop one scene, or play the whole arrangement.

16 · Tempo and key — beats per minute, then the song's key and scale.

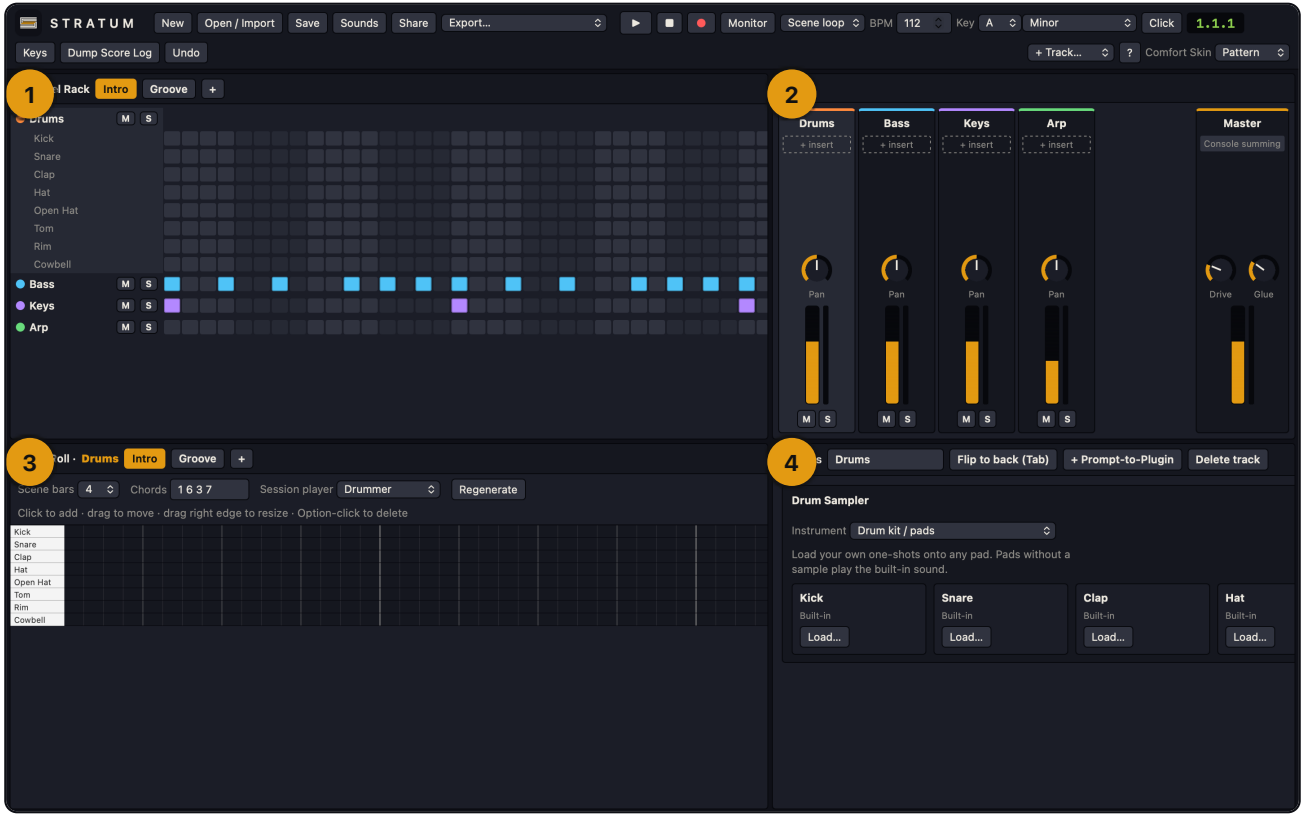
17 · Click — the metronome.

18 · Position — bar . beat . sixteenth of the playhead.

19 · + Track — add a synth, drum or audio track.

20 · ? — shortcuts for the current skin, the tour and this manual.

21 · Comfort Skin — choose the layout that matches the software you know (Chapter 3).



The panels in the Pattern skin.

1 · Channel Rack — every track, with a step grid for drum patterns.

2 · Mixer — a channel strip per track plus the master.

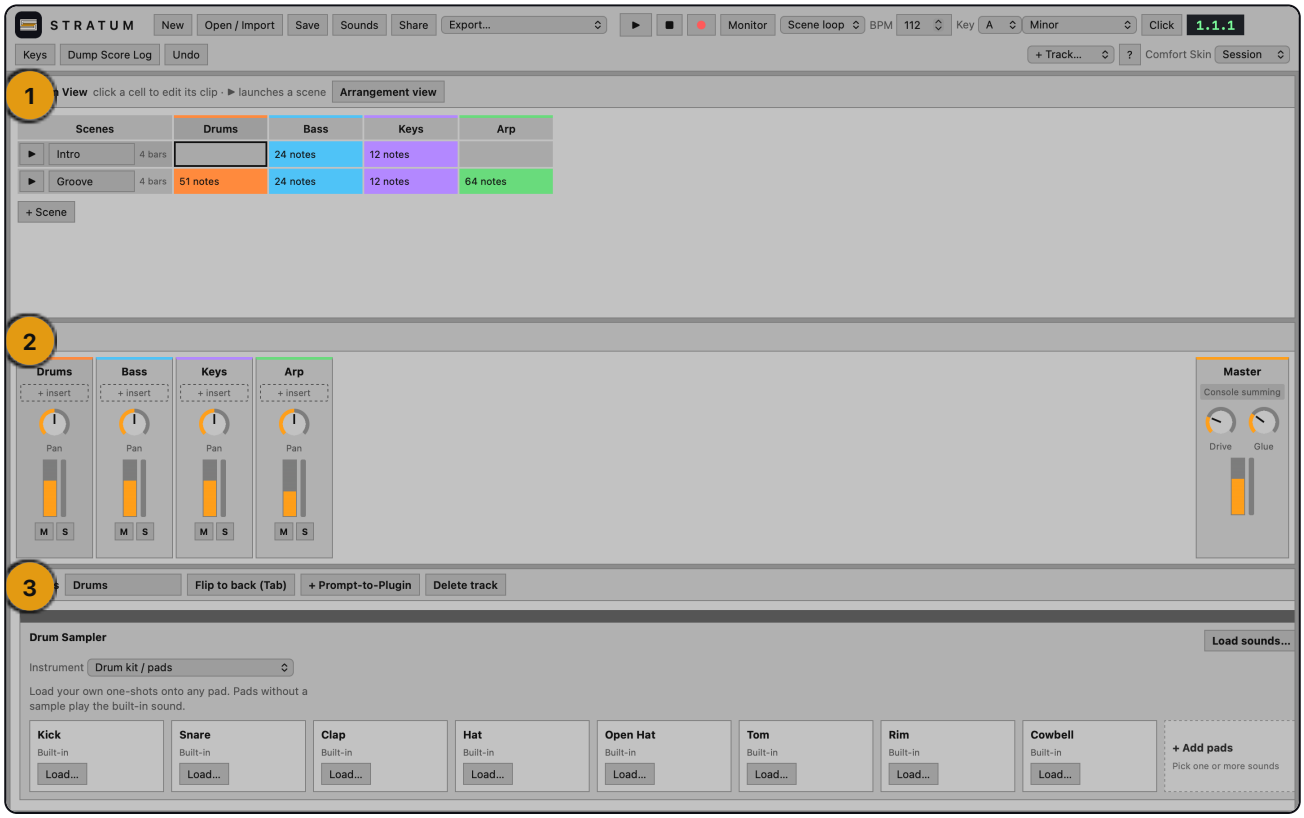
3 · Piano Roll — the notes of the selected clip.

4 · Devices — the selected track's instrument and effects.

3 Comfort Skins

A Comfort Skin rearranges the window and changes the keyboard shortcuts so STRATUM feels like the software you are used to. The project itself does not change — switch skins as often as you like from the **Comfort Skin** menu at the top right.

Skin	Feels like	What you see
Pattern	FL Studio	Channel Rack with step grid, Mixer, Piano Roll and Devices; F5 and F7 switch the lower panel to the Playlist or Piano Roll.
Session	Ableton Live	Scenes and clips in a grid, the Mixer, and clip or device view at the bottom (⇧Tab).
Arrange	Logic Pro, Cubase, Studio One	Inspector on the left, tracks and regions on a timeline, editor or mixer below.
Mix / Edit	Pro Tools	Separate Edit and Mix windows (⌘=).

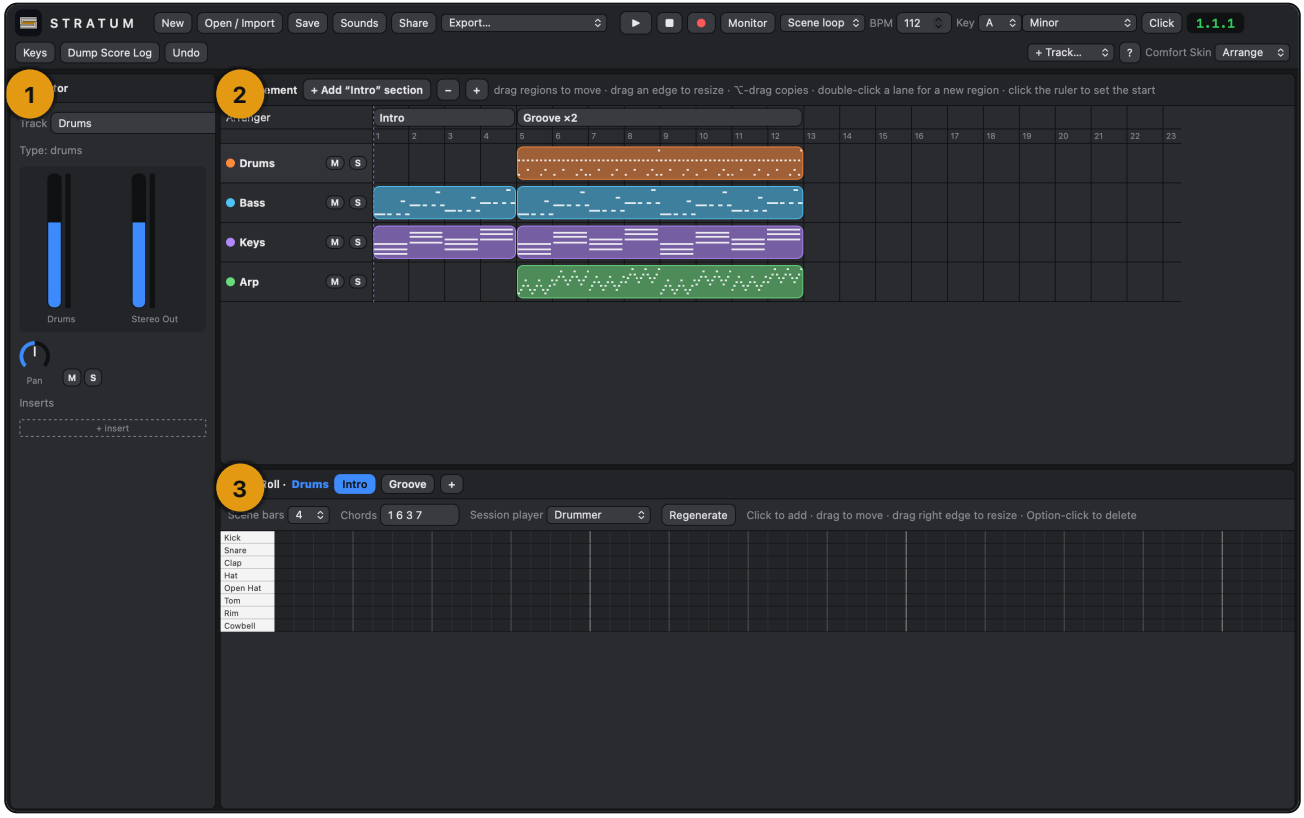


The Session skin.

1 · Clip grid — scenes in rows, tracks in columns; ▶ launches a whole scene.

2 · Mixer — levels, pan, mute and solo.

3 · Device view — instrument and effects of the selected track.

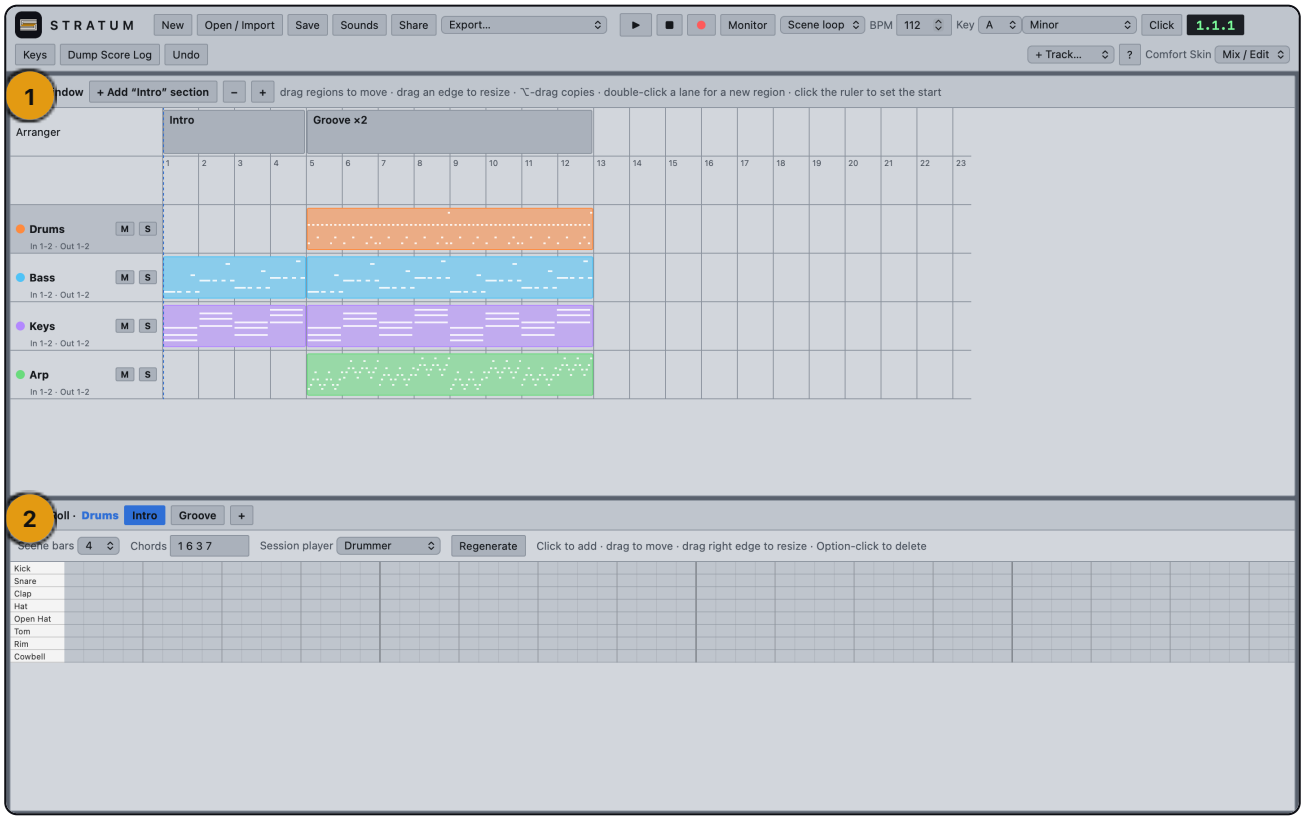


The Arrange skin.

1 · Inspector — the selected track's level, pan and inserts.

2 · Tracks area — regions on the song timeline, with section markers above.

3 · Editor — the piano roll for the selected region.



The Mix / Edit skin.

1 · Edit window — the timeline; ⌘= swaps to the Mix window.

2 · Editor — notes of the selected region.

4 Projects, opening and importing

Save writes a `.stratum` file that contains everything, including recorded and imported audio, so it can be sent to someone else as a single file. **Open / Import** (⌘O) opens STRATUM projects and brings in work from other software:

File	What comes in	What doesn't
FL Studio `.flp` or zipped loop package	Tempo, channels, patterns, notes, the playlist, volume and pan, sampler and audio-clip samples (found on your Mac, in the zip, or via **Missing samples**), volume and pan automation.	FL Studio's own instruments (FLEX, Sytrus, 3x Osc…) play on stand-in sounds; mixer effects; tempo in some FL 25 projects.
Logic Pro `.logicx` (or a zip of one)	Tempo, key, tracks by name, MIDI regions with notes in place, Quick Sampler samples with their root key, bounced audio.	Mixer levels, effects, automation; Logic's own synths.
Ableton Live `.als`	Tempo, tracks, MIDI clips and arrangement, volume and pan.	Devices, samples and automation.
Reason `.reason`	Tempo, tracks, note clips, audio-track recordings.	Rack devices and their samples (bounce them to audio in Reason first).
MIDI `.mid`	Every track's notes and the tempo.	—
Audio (WAV, AIFF, MP3, M4A, FLAC)	A new audio track, stretched to the song tempo when its own tempo is detected.	—

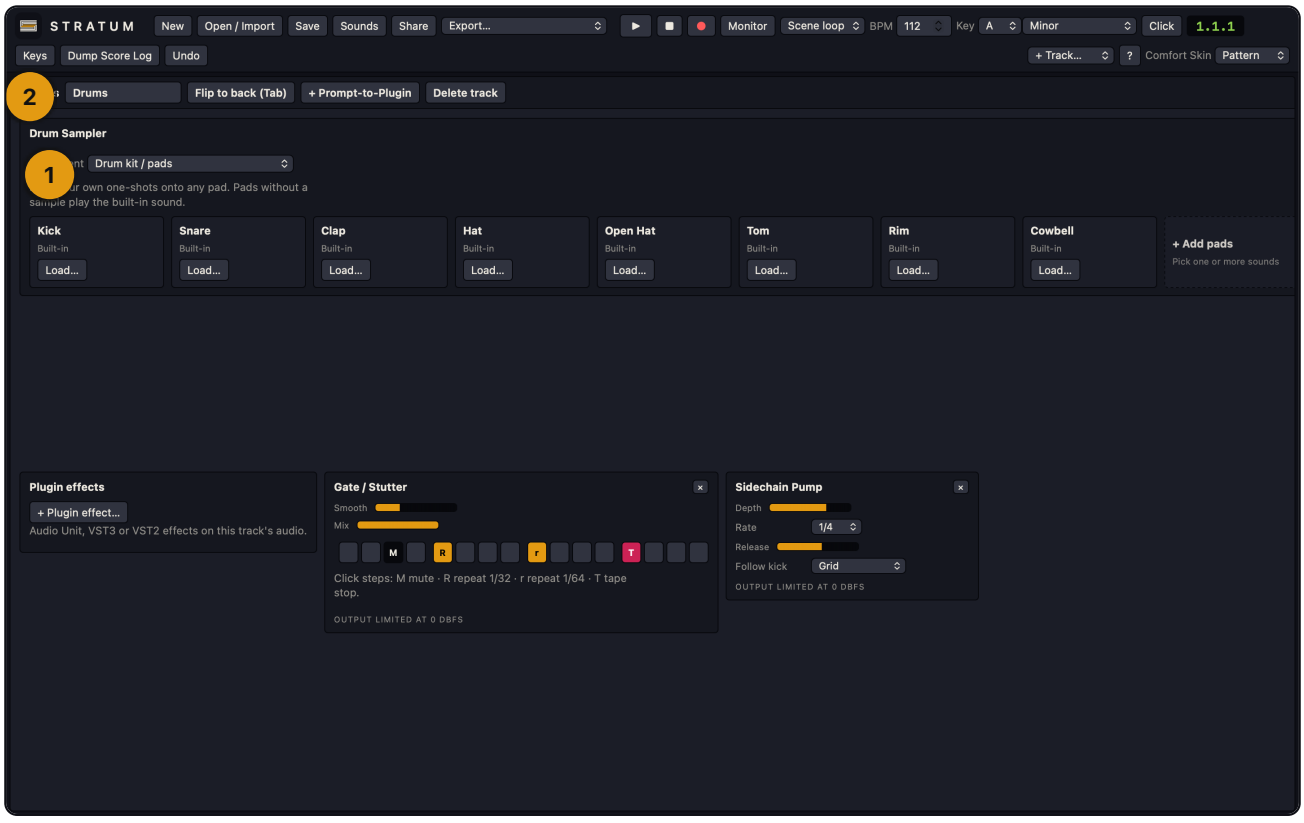
If a third-party plugin a project used is installed on your Mac, STRATUM puts it back on its track (with its default sound).

> Send STRATUM projects as `.stratum` files, and other people's FL Studio projects as zipped loop packages — those carry their samples.

5 Tracks and instruments

Add tracks with **+ Track**. Every note track has an **Instrument** menu at the top of its Devices panel, so the same notes can be played by any sound:

Instrument	What it is
Built-in synth	Two oscillators, a filter and an envelope. Quick and light.
Wavetable synth (STRATUM Wave)	STRATUM's full synthesizer — see Chapter 6.
My own sample…	Plays any sound file across the keys from its root note, with an optional loop.
Drum kit / pads	Eight pads with built-in sounds; load your own one-shots onto any pad, or add more pads.
Plugin…	Any Audio Unit, VST3 or VST2 instrument installed on your Mac (Chapter 17).



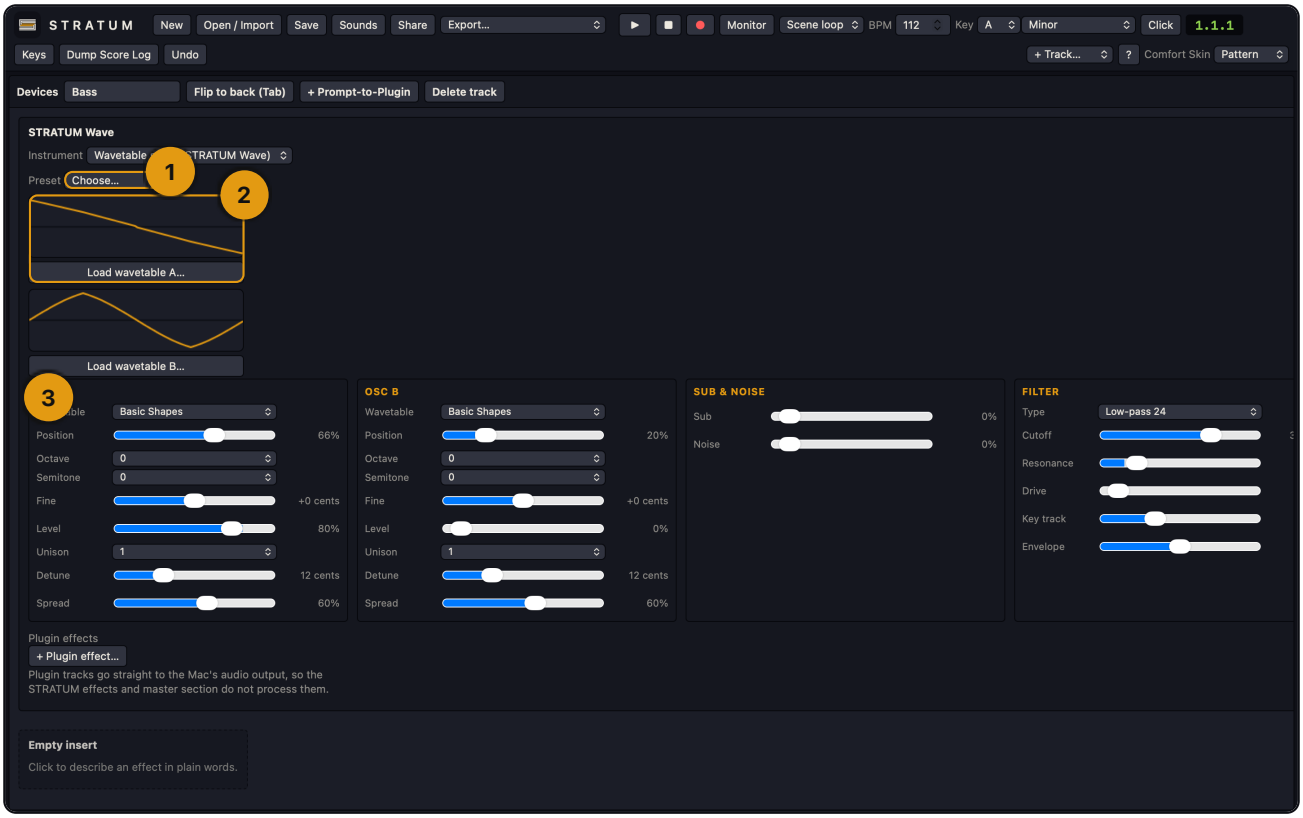
A drum track's Devices panel.

1 · Instrument menu — switch what this track plays; the notes stay the same.

2 · Drum Sampler — Load… puts your own sound on a pad; Load sounds… fills several pads at once, matching kicks, snares and hats by file name.

6 STRATUM Wave

STRATUM Wave is a wavetable synthesizer, in the same family as Serum, Vital and Pigments. A wavetable is a series of single waveforms; the **Position** control sweeps through them, which is what makes wavetable sounds move. Choose **Wavetable synth (STRATUM Wave)** in a track's Instrument menu.



STRATUM Wave in the Devices panel.

1 · Preset — ten starting points: Supersaw Lead, Reese Bass, Wobble Bass, Pluck, Warm Pad, Growl Bass and more.

2 · Waveform displays — the waveform each oscillator is playing right now; Load wavetable… turns any audio file into a wavetable.

3 · Controls — in sections: Osc A, Osc B, Sub & Noise, Filter, Amp Envelope, Mod Envelope, LFO and Voice.



Mod envelope and LFO can move the filter cutoff and the wavetable position; the LFO can follow the song tempo.

Section	Controls
Osc A / Osc B	Wavetable (nine built in, or your own), Position, Octave, Semitone, Fine, Level, Unison (1–8 voices), Detune, Spread (stereo width).
Sub & Noise	A sine an octave below, and white noise.

Filter	Off, low-pass 12 or 24, high-pass, band-pass; Cutoff, Resonance, Drive, Key track, Envelope amount.
Amp / Mod Envelope	Attack, Decay, Sustain, Release. The mod envelope can also move each oscillator's position.
LFO	Sine, triangle, saw, square or random; free rate or synced from 4 bars to 1/16 triplets; to cutoff, position, pitch (vibrato) and volume.
Voice	Poly, Mono or Legato, Glide, Velocity sensitivity, Volume.

> For a wobble bass, load the Wobble Bass preset and change **LFO › Sync** between 1/4, 1/8 and 1/16 to change the wobble speed in time with the song.

7 Writing notes and patterns

- **Step grid** (Pattern skin): click squares in the Channel Rack to place drum hits on sixteenth notes.
- **Piano Roll**: click to add a note, drag to move it, drag its right edge to change its length, Option-click to delete. **Scale lock** keeps new notes in the song's key.
- **Session players**: in the Piano Roll's Session player menu choose Drummer, Bass, Chords or Arpeggio and press Regenerate — STRATUM writes a part that follows the scene's chords (the Chords box, in scale degrees, e.g. 1 6 3 7).
- **Keys**: turn on Keys in the top bar to play the selected track from the computer keyboard (A–L row; Z and X change octave).
- **Dump Score Log**: STRATUM keeps the last five minutes of everything you played, recording or not. Press Dump Score Log to put it into the selected clip.

8 Arranging your song

STRATUM has two ways of playing, chosen with the **Scene loop / Song** menu:

- **Scene loop** repeats one scene — a pattern or a set of clips — while you work on it, like FL Studio's pattern mode or an Ableton scene.
- **Song** plays the arrangement: regions placed freely on each track's timeline, with named sections across the top.

In the timeline, drag regions to move them, drag their right edge to lengthen or loop them, ⌘E splits at the playhead and ⌘D duplicates. Section buttons move, duplicate or delete whole sections.

Automation: volume and pan automation imported from other software plays in Song mode; the Inspector shows a track's automation and can clear it. Drawing your own automation is planned for a later version.

9 Recording and comping

1. Add an audio track and press its arm button (or just press Record — STRATUM arms or adds an audio track for you).
2. Turn on **Monitor** if you want to hear yourself; this is direct monitoring, without effects.
3. Press **Record**. Each pass over the same place becomes a new **take**.
4. Comp the best parts: in the take lanes below a recorded region, drag across a take to use that part of it.

STRATUM removes the audio interface's delay from recordings automatically, so takes line up with the beat.

10 Effects

Effects are added in the Devices panel. STRATUM's own effects run in its audio engine; plugin effects run in the Mac's plugin engine.

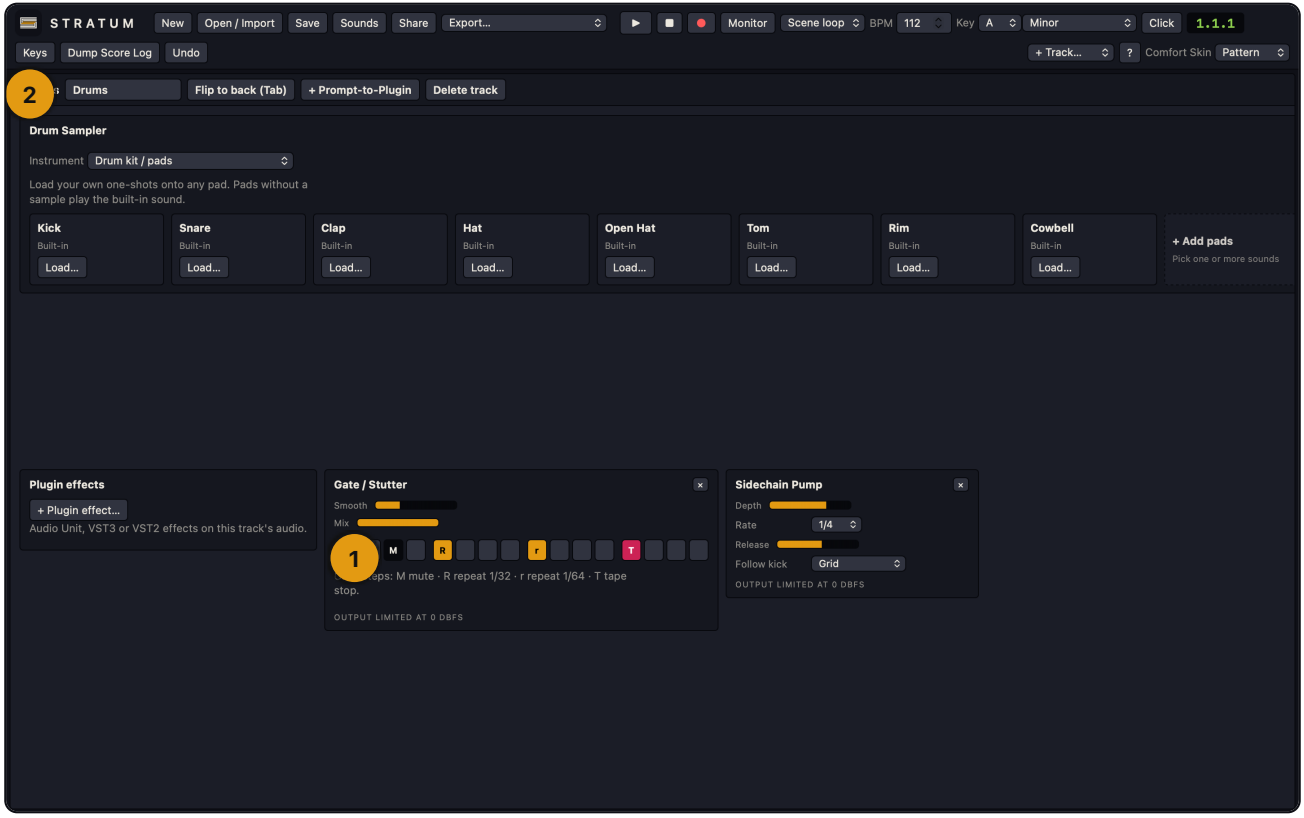


The signal path of a track. A track with any plugin on it — including STRATUM Wave, STRATUM Tune and Airwindows — plays through the plugin engine instead and skips STRATUM's own effects and the master section.

Prompt-to-Plugin

Click **+ Prompt-to-Plugin** and describe the effect you want in plain words — “dusty lo-fi tape with a little room”, “sidechain pump”, “stutter”. STRATUM builds a chain of effects to match and shows a knob for each setting.

Stock effect	Controls
High-pass / Low-pass	Cutoff (and Resonance)
Compressor	Amount
Saturation	Drive, Tone
Bit Crusher	Crush
Tape Wobble	Rate, Depth
Tone	Bright
Chorus / Phaser	Rate, Depth, Mix
Tremolo	Rate, Depth
Delay	Time, Feedback, Mix
Reverb	Size, Mix
Dust & Hiss	Level
Sidechain Pump	Depth, Rate (1 bar to 1/16), Release, Follow (grid or kick-drum hits)
Gate / Stutter	16-step pattern, Smooth, Mix



Gate / Stutter and Sidechain Pump on a drum track.

1 · Gate / Stutter steps — click a step to cycle: play, M mute, R repeat 1/32, r repeat 1/64, T tape stop.

2 · Sidechain Pump — ducks the track on the beat, or whenever a kick drum plays — a muted “ghost” kick track works too.

Airwindows

508 Airwindows effects are built in — reverbs, tape, saturation, console emulation, EQs, compressors, clippers and more. Choose **+ Plugin effect...**; Airwindows effects are listed with their category, and the recommended ones come first. Their controls appear in the Devices panel.

STRATUM Tune

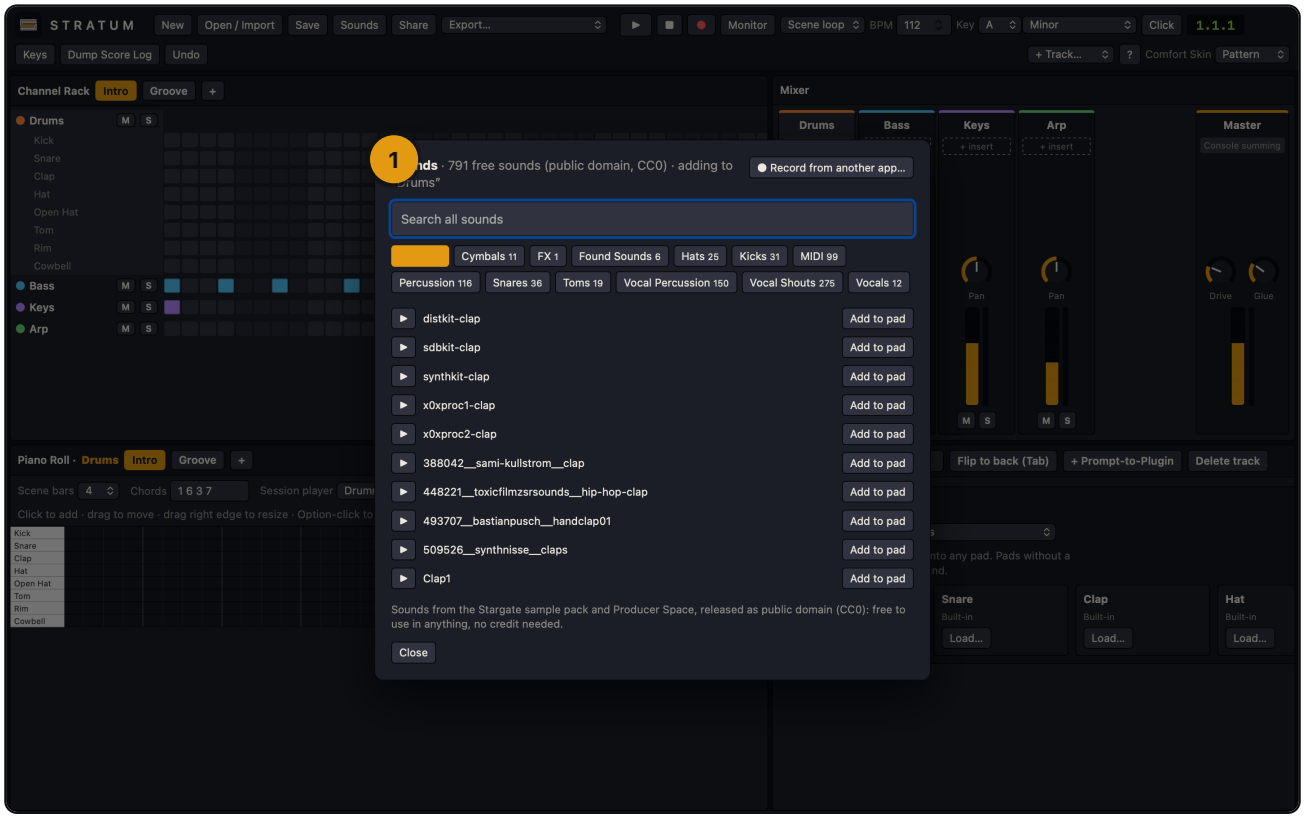
Pitch correction for vocals and other single-note parts. It starts in the song's key and scale; **Retune speed** runs from instant (the robotic effect) to natural, with **Amount** and **Mix**.

Back panel and LFOs

Press Tab to flip the Devices panel round. Patch an LFO's CV output into an effect's input to move that setting in time — like the back of a Reason rack.

11 The sound library

Click **Sounds** to browse 791 free sounds — kicks, snares, claps, hats, cymbals, toms, percussion, found sounds, vocal shouts and vocal percussion — and 99 MIDI chord, bass and piano kits. They are public domain (CC0): use them in anything, with no credit needed.



The Sounds browser.

1 · Sounds browser — search, pick a category, press ▶ to preview, and add the sound to the selected track: a drum pad, the sampler, an audio clip, or for MIDI, the notes.

Record from another app

Choose **File › Record from Another App...** or the button at the top of the Sounds browser. Pick the app (a web browser, a music player), choose where the sound goes — a new audio track, the sampler or a drum pad — press Record and play the sound in that app. Only that app is recorded, and silence at both ends is trimmed. The first time, macOS asks you to allow STRATUM under Privacy & Security › Screen & System Audio Recording. Only record sounds you have the right to use.

12 Mixing

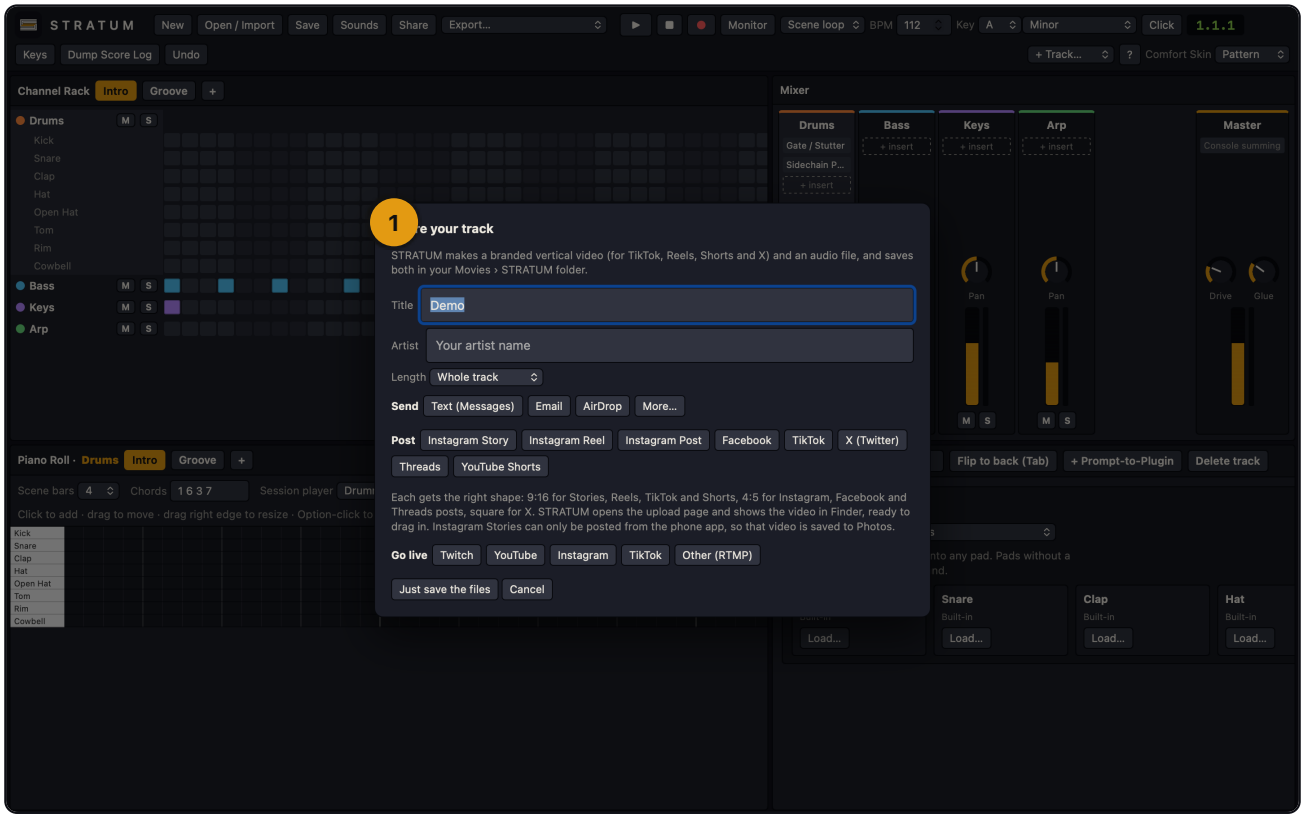
Each track has a volume fader, pan knob, mute (M) and solo (S). The **Master** strip has **Drive** (gentle console-style saturation) and **Glue** (bus compression). Each effect chain and the master end in a limiter, so STRATUM's own output doesn't clip.

13 Stems: splitting a track apart

Select an audio clip and press **Separate stems (AI)** in its clip settings. STRATUM separates the audio into drums, bass, vocals and other instruments on four new tracks, using the HT-Demucs model on your Mac — nothing is uploaded. This needs an Apple silicon Mac and can take about as long as the audio itself.

14 Exporting and sharing

Export saves a WAV mixdown, a MIDI file, or a DAWproject file for other DAWs. **Share** makes a STRATUM-branded video and an audio file of the whole track or its first 15, 30 or 60 seconds, saves them in **Movies › STRATUM**, and sends them:



The Share window.

1 · Share — title and artist, length, then where to send it.

Destination	Video shape	How
Messages, AirDrop	9:16	opens with the video attached
Mail	audio file	opens with the audio attached
Instagram Story	9:16, up to 60 s	saved to Photos to post from your phone
Instagram Reel, TikTok, YouTube Shorts	9:16	opens the upload page; the video is selected in Finder
Instagram Post, Facebook, Threads	4:5	opens the upload page
X (Twitter)	square, up to 2:20	opens the post composer

Going live

The **Go live** row in the Share window streams your session as you work. Viewers see the STRATUM window and, if you tick **Show my camera in a box**, your camera in a small box beside the window (the window shrinks a little to make room, so nothing is covered). They hear whatever your Mac plays, plus your microphone if you add it. Pick a platform, paste the server URL and stream key it gives you, and press **Go live**. A red ● **LIVE** timer appears at the top of the window. Press it to end the stream.

Platform	Where to find the server URL and stream key
Twitch	Creator Dashboard › Settings › Stream (STRATUM fills in the server)
YouTube	YouTube Studio › Create › Go live › Stream (STRATUM fills in the server)
Instagram	instagram.com on a computer › Create › Live video; press Go live in Instagram once STRATUM is streaming
TikTok	LIVE Center or TikTok LIVE Studio, for accounts with LIVE access
Other (RTMP)	any service that gives a server URL and stream key: Kick, Facebook Live, Restream…

Tick **Remember for next time** to keep the key in your Mac's Keychain. Keep your stream key private: anyone who has it can stream to your channel. If you stream with OBS today, STRATUM does that job on its own, sending 720p video at 30 frames a second.

15 Coming from another DAW

Most ideas you know have a home in STRATUM. Choose the matching Comfort Skin first, then use this table to find things.

You know...	In STRATUM
FL Studio Channel Rack, step sequencer	Pattern skin › Channel Rack step grid
FL Studio Playlist	Song mode timeline (F5)
FL Studio patterns	Scenes
FL Studio mixer inserts	Devices panel effects
FL Studio Fruity Edison / slicing	Stems and the sampler
Ableton Live Session View	Session skin clip grid
Ableton Arrangement View	Song mode timeline
Ableton devices and racks	Devices panel; Prompt-to-Plugin chains
Ableton Simpler	My own sample / drum pads
Ableton Wavetable	STRATUM Wave
Logic Pro Tracks area, regions	Arrange skin
Logic Quick Sampler, Drum Machine Designer	My own sample, Drum kit / pads
Logic Alchemy, Vital, Serum	STRATUM Wave
Logic Flex Pitch / Auto-Tune	STRATUM Tune
Logic Step Sequencer	Pattern skin step grid; Gate / Stutter
Pro Tools Edit and Mix windows	Mix / Edit skin (⌘=)
Pro Tools playlists and comping	Takes and comp lanes
Reason rack and back panel cables	Devices panel; Tab flips it to patch LFOs
Reason Kong	Drum kit / pads
Reason Europa	STRATUM Wave
GarageBand Smart Controls	Device knobs and plugin controls
Kickstart, LFO Tool	Sidechain Pump
Gross Beat, Effectrix	Gate / Stutter

16 Keyboard shortcuts

Shortcuts change with the Comfort Skin so they match the software it imitates. Press **?** in STRATUM to see the current set.

Every skin

Key	Does
`Space`	Play / stop
`Tab`	Flip the rack to its back panel
`⌘ z`	Undo
`⌘ s`	Save project
`⌘ o`	Open / import
`Backspace`	Delete the selected note or region
`⌘ e`	Split region at the playhead
`⌘ d`	Duplicate region or scene

Pattern

Key	Does
`l`	Pattern / Song mode
`F5`	Playlist
`F7`	Piano roll
`=`	Next pattern
`-`	Previous pattern
`⌘ b`	Duplicate pattern
`r`	Record

Session

Key	Does
`⇧ Tab`	Clip view / device view
`⌘ ⇧ t`	Insert MIDI track
`⌘ t`	Insert audio track
`⌘ i`	Insert scene

`⌘ d`	Duplicate scene
`F9`	Record
`F10`	Session / Arrangement view

Arrange

Key	Does
`Enter`	Go to beginning
`p`	Piano roll
`x`	Mixer
`b`	Devices
`k`	Metronome
`r`	Record
`⌘ t`	Split region at the playhead
`m`	Mute track
`s`	Solo track
`⌘ ⌥ s`	New software instrument track
`⌘ ⌥ a`	New audio track

Mix / Edit

Key	Does
`Enter`	Return to zero
`⌘ =`	Toggle Mix / Edit windows
`⇧ m`	Mute track
`⇧ s`	Solo track
`⌘ ⇧ n`	New track
`7`	Click on / off
`F12`	Record

17 Plugins

STRATUM finds the Audio Unit, VST3 and VST2 plugins installed on your Mac. Choose **Plugin…** in a track's Instrument menu for an instrument, or **+ Plugin effect…** for an effect on any track.

- **Open plugin window** shows the plugin's own controls; **Controls** shows its settings inside STRATUM's Devices panel.
- Plugin settings are saved in the project and included in exports.
- Plugins receive the song's tempo and position, so tempo-synced plugins follow the song.
- VST3 plugins are looked for in `/Library/Audio/Plug-Ins/VST3` and `~/Library/Audio/Plug-Ins/VST3`; VST2 in the matching `VST` folders.

> Plugins run inside STRATUM, so a faulty plugin can close the app. Save often when trying a new plugin.

18 Limits in this alpha and troubleshooting

- **“STRATUM can't be opened”** — use System Settings › Privacy & Security › Open Anyway (Chapter 1).
- Stem separation needs an Apple silicon Mac.
- Many older VST2 plugins are Intel-only and won't load on Apple silicon.
- Imported projects bring notes, arrangement and samples, but not other software's own instruments, mixer effects or most automation.
- STRATUM can't post to social platforms by itself; it opens each upload page with your video ready to drag in.
- Automation can be played and cleared but not yet drawn.

Licences

Airwindows Consolidated (MIT) · Steinberg VST3 SDK (MIT) · ONNX Runtime (MIT) · HT-Demucs (MIT) · Stargate sample pack and Producer Space sounds (CC0). Licence files are inside the app.

19 Windows, phones and tablets

STRATUM also runs on Windows and, as a web app, on iPhone, iPad, Android phones and tablets. Projects are the same `.stratum` files everywhere, so you can start a beat on your phone and finish it on the Mac.

Windows

Unzip **STRATUM-Windows.zip**, keep **STRATUM.exe** next to the **app** folder, and double-click it. If Windows says it protected your PC, choose **More info › Run anyway**. Shortcuts use **Ctrl** and **Alt** where the Mac uses ⌘ and ⌥, and the tour is under the **?** button.

Phones and tablets

Open **irlewis91.github.io/stratum** in Safari (iPhone, iPad) or Chrome (Android). Tap **Share › Add to Home Screen** on iPhone and iPad, or **Install app** on Android. STRATUM then opens full screen from its icon and works offline. On a phone, playback controls sit on the top row, the other tools scroll sideways underneath, and a tab bar along the bottom switches between the Rack, Piano roll, Mixer and Devices. Tap a note, then **Delete**, to remove it. On iPhone, turn the ringer on: iOS mutes web audio in silent mode.

Moving a beat between devices

A beat is one `.stratum` file, so it opens anywhere STRATUM runs. On the Mac, save it in iCloud Drive or AirDrop it. On Windows, **Save** offers **Save to a folder...** (pick your OneDrive, Google Drive or Dropbox folder) and **Share...**. On a phone or tablet, **Save** opens the share sheet (AirDrop, Save to Files, iCloud Drive, Google Drive, email). Then press **Open / Import** on the other device and choose the file. There is no automatic syncing yet.

	Mac	Windows	Phones, tablets, browsers
Beat making, piano roll, mixer, skins, built-in sounds and effects	✓	✓	✓
Sound library	✓	✓	✓ (sounds download as you use them)
Importing projects from other software	✓	✓	✓
Recording from a microphone	✓	✓	✓
Branded share videos	✓	✓	✓ through the share sheet
Plugins, STRATUM Wave, STRATUM Tune, Airwindows	✓	—	—
Stems, Record from another app, Go live	✓	—	—

Share videos on Windows, phones and tablets are recorded as the track plays once, so a 60-second clip takes about a minute to make.